

# JEE Main - 15 | JEE - 2022

Date: 27/11/2021

Maximum Marks: 300

Timing: 04:00 PM to 09:00 PM

## General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the  $\oplus$  sign for positive values. However, for negative values,  $\ominus$  sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

## Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : .....

Roll Number : .....

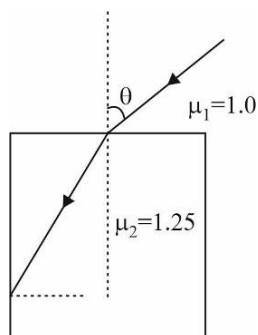
OMR Bar Code Number : .....

Candidate's Signature : ..... Invigilator's Signature .....

**PART I : PHYSICS****100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

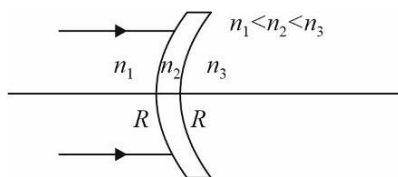
1. Consider the situation as shown in the figure. What will be the maximum angle  $\theta$  for which the light suffers total internal reflection at the vertical surface?



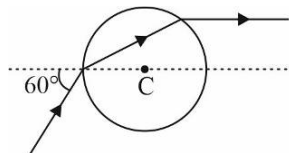
- (A)  $\sin^{-1}\left(\frac{4}{5}\right)$       (B)  $\sin^{-1}\left(\frac{3}{4}\right)$       (C)  $\sin^{-1}\left(\frac{3}{5}\right)$       (D)  $\sin^{-1}\left(\frac{1}{2}\right)$
2. An object of height +2.5 cm is placed perpendicular to the principal axis of a concave mirror of focal length  $-x$  cm, at a distance of  $-1.5x$  cm from the pole of the mirror. The height of the image will be : ( $x > 0$ )
- (A) +1.5 cm      (B) +3.0 cm      (C) -7.5 cm      (D) -5.0 cm
3. A hydrogen atom moving at a speed  $V$  metre/second absorbs a photon of wavelength 122 nm and stops. The value of  $V$  will be: ( $m_p = 1.67 \times 10^{-27}$  kg,  $h = 6.63 \times 10^{-34}$  J-s)
- (A) 1.67      (B) 3.25      (C) 5.43      (D) 25
4. Monochromatic radiation of wavelength  $\lambda$  is incident on a hydrogen sample in ground state. Hydrogen atoms absorb a fraction of light and subsequently emit radiation of six different wavelengths. The wavelength  $\lambda$  will be: ( $hc = 1242$  eV-nm)
- (A) 420 nm      (B) 210 nm      (C) 97.5 nm      (D) 41.25
5. An X-ray tube operates at 40 kV. A particular electron losses 2.5% of its kinetic energy to emit an X-ray photon at the first collision. The wavelength corresponding to this photon will be : ( $hc = 1242$  eV-nm)
- (A) 12.42 nm      (B) 7.2 nm      (C) 120 nm      (D) 1.24 nm
6. The width of one of the two slits in a Young's double slit experiments is 3 times of the other slit. Assuming that the amplitude of the light coming from a slit is proportional to the slit-width. The ratio of the maximum to the minimum intensity in the interference pattern will be:
- (A)  $\frac{3}{2}$       (B) 3      (C) 4      (D) 9
7. A beam of light consisting of two wavelengths,  $6500\text{\AA}$  and  $5200\text{\AA}$  is used to obtain interference fringes in a Young's double slit experiment. The distance between the slits is 2.0 mm and the distance between the plane of the slits and the screen is 120 cm. the least distance from the central maximum where the bright fringes due to both wavelengths coincide will be:
- (A) 2.44 mm      (B) 12.25 mm      (C) 1.56 mm      (D) 16.50 mm

8. A rod of length 10 cm lies along the principal axis of a concave mirror of focal length 10 cm in such a way that the end closer to the pole is 20 cm away from it. The length of the image will be :  
 (A) 10 cm (B) 15 cm (C) 12 cm (D) 5 cm
9. The nucleus of  $^{16}_8\text{O}$  has 8 protons and 8 neutrons. The binding energy per nucleon for oxygen atom will be: ( $m_p = 1.00727u$ ;  $m_e = 0.00055u$ ;  $m_n = 1.00866u$ ; Actual mass of  $^{16}_8\text{O}$  atom =  $15.99493u$ ;  $1u = 931.5\text{ MeV}$ )  
 (A) 7.97 MeV (B) 6.58 MeV (C) 9.25 MeV (D) 4.22 MeV
10. The wavelength of the first spectral line in Balmer series of hydrogen atom is  $6561\text{\AA}$ . the wavelength of the second spectral line in the Balmer series of doubly ionized Lithium atom will be :  
 (A)  $540\text{\AA}$  (B)  $730\text{\AA}$  (C)  $2570\text{\AA}$  (D)  $1460\text{\AA}$
11. A proton and an  $\alpha$ -particle are accelerated from rest by potential difference  $V$  and  $2V$  respectively. After this their de-Broglie wavelength are  $\lambda_p$  and  $\lambda_\alpha$  respectively. The ratio  $\frac{\lambda_\alpha}{\lambda_p}$  will be :  
 (A)  $\frac{1}{16}$  (B) 8 (C)  $\frac{1}{4}$  (D) 2
12. A thin lens (made of glass,  $\mu_{\text{glass}} = 1.5$ ) of focal length  $+12\text{ cm}$  is immersed in water of refractive index 1.32. The new focal length in the water will be :  
 (A) 32 cm (B) 44 cm (C) 18 cm (D) 9 cm
13. A radioactive material decays by simultaneous emission of two particles with respective half-lives 1620 and 405 year. The time, in years, after which one-fourth of the material remains is:  
 (A) 1620 (B) 1080 (C) 548 (D) 648
14. A parallel beam of monochromatic light of wavelength  $663\text{ nm}$  is incident on a totally reflecting plane mirror. The angle of incidence is  $37^\circ$  and the number of photons striking the mirror per second is  $3.5 \times 10^{21}$ . The force exerted by light beam on the mirror will be:  
 ( $\tan 37^\circ = 0.75$ ; value of Planck constant is  $6.63 \times 10^{-34}\text{ J-s}$ )  
 (A)  $4.2 \times 10^{-6}\text{ N}$  (B)  $3.6 \times 10^{-6}\text{ N}$  (C)  $5.6 \times 10^{-4}\text{ N}$  (D)  $2.4 \times 10^{-4}\text{ N}$
15. White light is used to illuminate the two slits in Young's double slit experiment. The separation between the slits is  $d$  and the screen is at a distance  $D(>d)$  from the slits. At a point on the screen directly in front of one of the slits certain wavelengths are missing. Which one of the given wavelength is missing:  
 (A)  $\lambda = \frac{d^2}{D}$  (B)  $\lambda = \frac{2d^2}{D}$  (C)  $\lambda = \frac{d^2}{2D}$  (D)  $\lambda = \frac{4d^2}{D}$
16. A beaker containing liquid is placed on a table underneath a microscope which can be moved along a vertical scale. The microscope is focused, through the liquid, on a mark on the table and the reading on the scale is  $r_1$ . It is next focused on the upper surface of the liquid and the reading is  $r_2$ . More liquid is added and the observation are repeated, the corresponding readings being  $r_3$  and  $r_4$ . The refractive index of the liquid is:  
 (A)  $\frac{r_4 - r_3 - r_2 + r_1}{r_4 - r_2}$  (B)  $\frac{r_2 - r_4}{r_4 - r_2 - r_3 + r_1}$  (C)  $\frac{r_4 - r_2}{r_4 - r_3 - r_2 + r_1}$  (D)  $\frac{r_4 - r_2}{r_1 + r_2 - r_3 - r_4}$

17. The focal length of the thin lens shown in the figure will be : (The radii of curvature of both the surfaces are equal to  $R$ ;  $n_1$ ,  $n_2$  and  $n_3$  are respective refractive index of the medium)



- (A)  $\frac{2(n_3 - n_1)R}{n_2}$  (B)  $\frac{n_3 R}{n_3 - n_1}$  (C)  $\frac{(n_1 + n_3)R}{n_2}$  (D)  $\frac{(n_1 + n_3)R}{(n_2 - n_1)}$
18. A light ray is incident on a transparent sphere making an angle of  $60^\circ$  with the diameter such that the emergent ray becomes parallel to the diameter as shown. Refractive index of the material of the sphere is:



- (A)  $\sqrt{5}$  (B) 2 (C)  $\sqrt{3}$  (D)  $\sqrt{2}$
19. When an object is at distances of  $u_1$  and  $u_2$  from the pole of a concave mirror, images of the same size are formed. Focal length of the mirror will be:

- (A)  $\frac{u_1 + u_2}{2}$  (B)  $\frac{u_1 u_2}{u_1 + u_2}$  (C)  $\frac{u_1^2}{2u_2}$  (D)  $\frac{u_2^2}{2u_1}$

20. A double slit experiment is performed with light of wavelength 576 nm and the interference pattern is observed on a screen 1m away. The tenth bright fringe has its centre at a distance of 1.2 cm from the central maximum. The separation between the slits is:

- (A) 1.24 mm (B) 1.44 mm (C) 0.48 mm (D) 2.42 mm

SPACE FOR ROUGH WORK

**SECTION-2**

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the  $\oplus$  sign for positive values. However, for negative values,  $\ominus$  sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. A point source S is placed at a distance of 15 cm from a converging lens of focal length 10 cm on its principal axis. A diverging mirror of focal length 12 cm is placed at a distance X cm from the lens such that a real image is formed on the source itself. The value of X is:
22. A ray of light is incident normally on one face of an isosceles prism that after suffering two total internal reflections, it emerges normally through base. The vertex angle (in degree) of the prism is \_\_\_\_\_.
23. Half-life of a radioactive substance A is half of the half-life of another radioactive substance B. Initially the number of nuclei of A and B are  $N_A$  and  $N_B$  respectively. After 3 half-lives of B, number of nuclei of both are equal. Then the ratio of  $\frac{N_A}{N_B}$  will be:
24. At two points A and B on a screen, in Young's double slit experiment, waves from slits  $S_1$  and  $S_2$  have a path difference of  $\lambda$  and  $\frac{\lambda}{4}$  respectively ( $\lambda$  is the wavelength of the light used). The ratio of intensities at A and B is  $(k+1)$ . The value of k is \_\_\_\_\_.
25. In a Young's double slit experiment, light of wavelength 600 nm is used. One of the slits is covered by a transparent sheet of thickness  $1.8 \times 10^{-5} m$  made of a material of refractive index 1.5.  $3N$  fringes are shifted due to the introduction of the sheet. The value of N will be:

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**SPACE FOR ROUGH WORK**

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**PART II : CHEMISTRY****100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Which of the following is NOT true for pnictogen family?
  - Tendency to exhibit  $-3$  oxidation state decreases down the group
  - The stability of  $+5$  oxidation state decreases down the group
  - In all elements all intermediate oxidation states disproportionate into  $+5$  and  $-3$  in alkali and acid
  - Nitrogen does not form compounds in  $+5$  oxidation state with halogens
- In chemistry a one-pot synthesis is a strategy to improve the efficiency of a chemical reaction where by a reactant is subjected to successive chemical reactions in just one reactor. This is much desired by chemists because avoiding a lengthy separation process and purification of the intermediate chemical compounds can save time and resources while increasing chemical yield. The Wolff-Kishner reduction reaction may be performed as a one-pot-reaction. Which of the following is correct for Wolff-Kishner reduction reaction.
  - It requires highly basic conditions
  - It is not suitable for base-sensitive substrates
  - Formation of intermediate hydrazone will not occur at sterically hindered carbonyl groups, preventing the reaction
  - This is superior to the related reduction reaction Clemmensen reduction for compounds containing acid sensitive functional group

(A) Only I, II      (B) Only III, IV      (C) I, II, III, IV      (D) None of these
- Which of the following is not matched correctly ?
 

|     |               |   |                           |
|-----|---------------|---|---------------------------|
| (A) | Phenylacetone | : | 1-benzylethan-1-one       |
| (B) | Acetophenone  | : | 1-phenylethan-1-one       |
| (C) | Propiophenone | : | 1-phenylpropan-1-one      |
| (D) | Benzophenone  | : | 1, 1-diphenylmethan-1-one |
- Which of the following is correct for oxoacids  $\text{H}_4\text{P}_2\text{O}_6$  and  $\text{H}_4\text{P}_2\text{O}_7$  ?
 

|     |                                                 |     |                                                       |
|-----|-------------------------------------------------|-----|-------------------------------------------------------|
| (A) | Both are tetrabasic                             | (B) | Both are reducing                                     |
| (C) | Both contains $\text{P}-\text{O}-\text{P}$ bond | (D) | Both can be prepared from red $\text{P}_4$ and alkali |
- Which of the following reaction involve formation of maximum number of paramagnetic species ?
 

|     |                                          |     |                                      |
|-----|------------------------------------------|-----|--------------------------------------|
| (A) | $\text{NO} + \text{O}_3 \longrightarrow$ | (B) | $\text{NaNO}_3 \xrightarrow{\Delta}$ |
| (C) | $\text{NaN}_3 \xrightarrow{\Delta}$      | (D) | $\text{KClO}_3 \xrightarrow{\Delta}$ |
- $$\text{Cl}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{CH}_2-\underset{\text{CHO}}{\underset{|}{\text{CH}}}-\text{CH}=\text{CH}_2 \xrightarrow[4. \text{PCC}]{1. \text{H}_2, \text{Pd}-\text{BaSO}_4, 2. \text{B}_2\text{H}_6/\text{THF}, 3. \text{H}_2\text{O}_2/\text{HO}^-} (\text{X})$$

Which of the following is correct IUPAC name of organic compound X ?

|     |                                 |     |                                 |
|-----|---------------------------------|-----|---------------------------------|
| (A) | 3-carbaldehydepentan-1, 5-diol  | (B) | 3-ketopentan-1, 5-dial          |
| (C) | Propane-1, 2, 3-tricarbaldehyde | (D) | 3-hydroxymethylpentan-1, 5-dial |

7. Consider the reaction I and II.



Which of the following is correct ?

- (A) Both are third order (overall) reactions (B) Both are non redox reactions  
(C) Both produce same product(s) (D) Both follows same mechanism

8. Which of the following reaction required charcoal as catalyst?

- (A)  $\text{SO}_2 + \text{O}_2 \longrightarrow \text{SO}_3$  (B)  $\text{SO}_2 + \text{Cl}_2 \longrightarrow \text{SO}_2\text{Cl}_2$   
(C)  $\text{P}_4 + \text{O}_2 \longrightarrow \text{P}_4\text{O}_{10}$  (D)  $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$

9. Camphor (X) is a waxy, flammable, transparent solid with strong aroma.



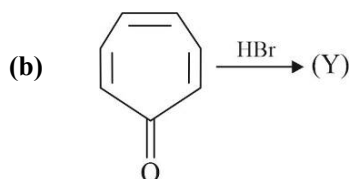
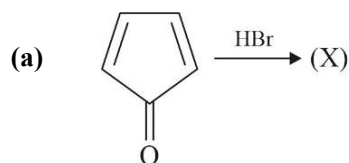
Which of the following is correct ?

- (a) Its degree of unsaturation is two  
(b) It produce mixture of monocarboxylic acids having lesser number of carbon atoms than the parent ketone, on vigorous oxidation  
(c) It contains two chiral carbon atoms  
(d) It has only two possible enantiomers  
(A) a, c (B) a, b, c (C) only c (D) c, d

10. The standard electrode potential of halogen decreases down the group. Its value is independent of :

- (A) Bond dissociation enthalpy (B) Electron gain enthalpy  
(C) Electronegativity (D) Hydration enthalpy

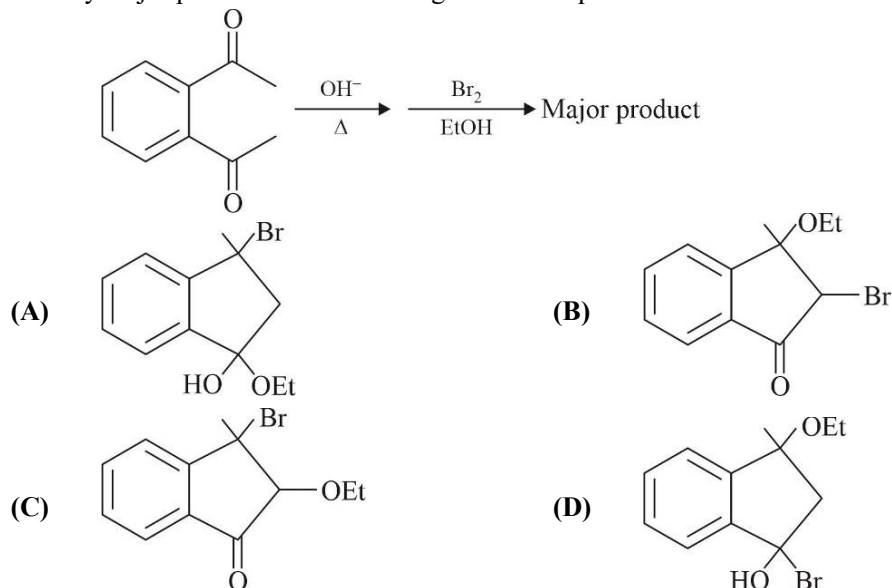
11. Consider the reactions a and b.



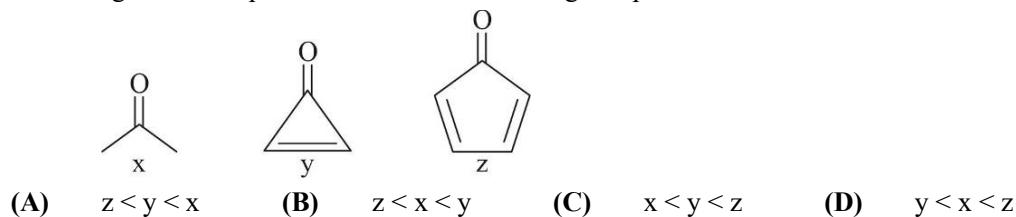
Identify correct statement :

- (A) Both (X) and (Y) are soluble in organic solvent  
(B) Both (X) and (Y) are soluble in water  
(C) (X) is soluble in organic solvent while (Y) is soluble in water  
(D) (X) is soluble in water while (Y) is soluble in organic solvent

12. Identify major product of the following reaction sequence.



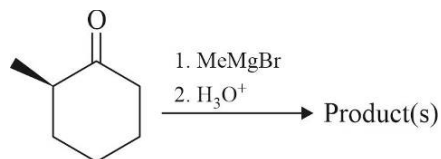
13. Increasing order of dipole moment of the following compound.



14. Which of the following is used for the production of  $\text{UF}_6$  in the enrichment of  $^{235}\text{U}$ ?

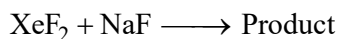


15. What is stereochemical outcome of this reaction?



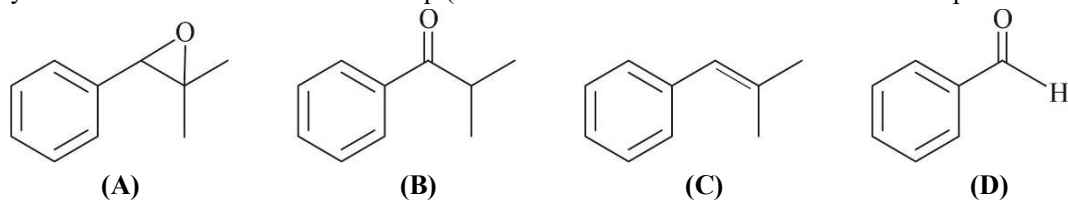
- (A)** Racemic mixture      **(B)** A single achiral molecule  
**(C)** Mixture of diastereomers      **(D)** An unequal mixture of enantiomers

16. What is the shape of anionic species of product in the following reaction?



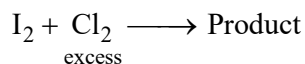
- (A)** Triangular planar      **(B)** Pyramidal  
**(C)** T-shape      **(D)** Octahedral

17. You are trying to make the alcohol, 2-methyl-1-phenylpropan-1-ol, from which of these structures can you NOT make the alcohol in one step (reaction and acidification or oxidation workup = one step)

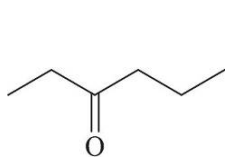




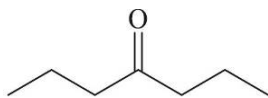
18. Which of the following is not true regarding product of the following reaction ?



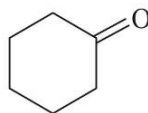
- (A) Product is orange solid  
 (B) Product can exist in planar dimeric form  
 (C) Product can be hydrolyzed  
 (D) Structure of product is pentagonal bipyramidal
19. Alkyne hydration would be a good preparative method for one of the following compounds. Identify compound which can be prepared from alkyne hydration in high yield.



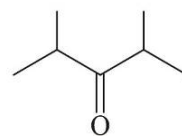
(A)



(B)

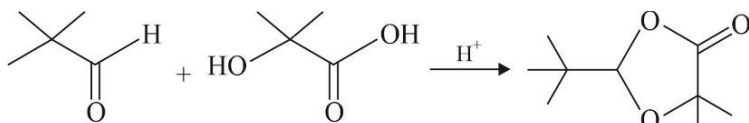


(C)



(D)

20. Consider the following reaction and identify true statement.



- (A) It involve hemiacetal formation and intramolecular esterification  
 (B) It involve acetal formation and intra molecular esterification  
 (C) It involve intermolecular esterification and acetal formation  
 (D) It involve inter molecular esterification and hemiacetal formation

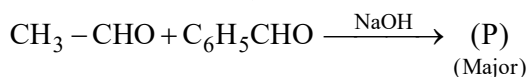
SPACE FOR ROUGH WORK

**SECTION-2**

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the  $\oplus$  sign for positive values. However, for negative values,  $\ominus$  sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

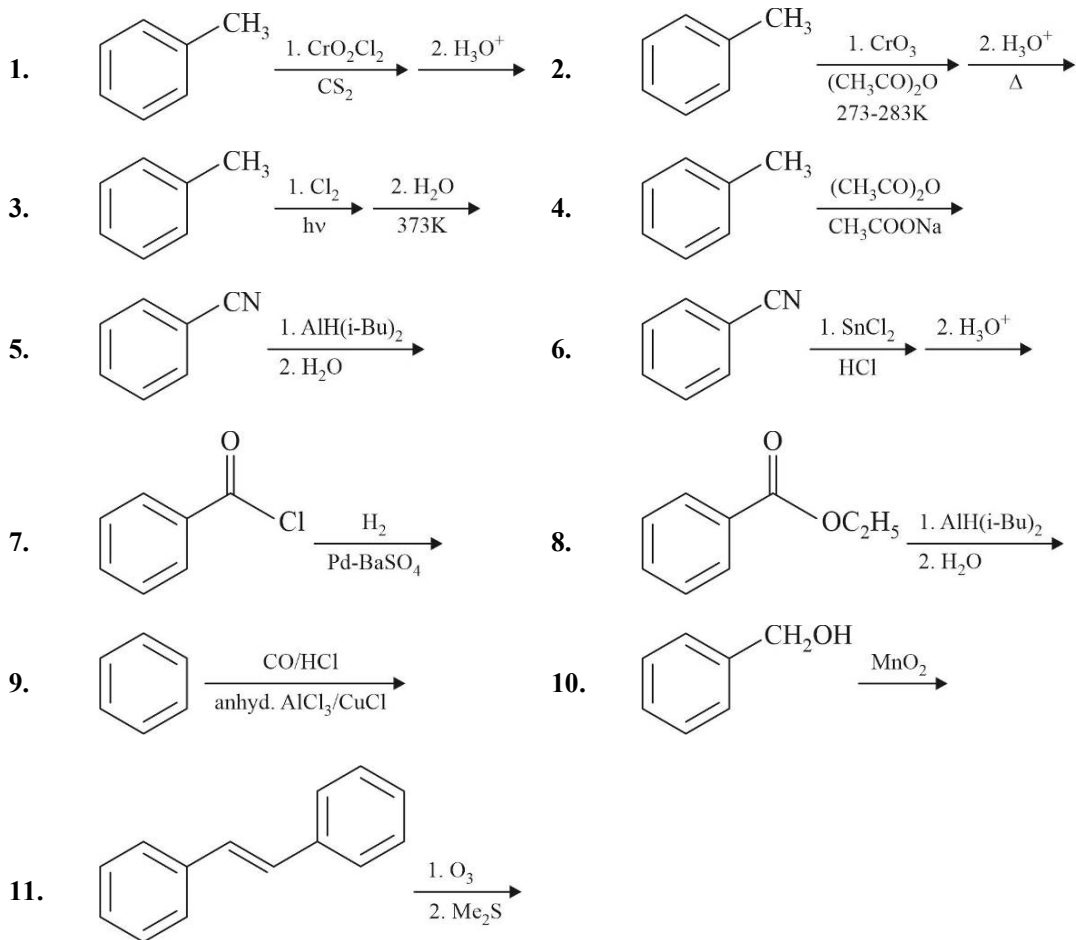
21. Find the sum of number of  $sp^3$  hybridized atoms and lone pair electrons in mustard gas.

22. Consider the following reaction.

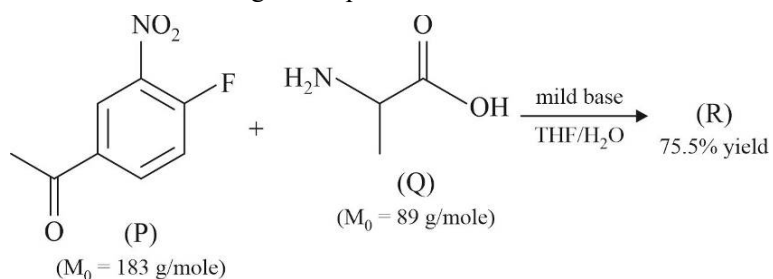


Find the number of  $sp^2$  hybrid orbitals in most stable form of product (P).

23. How many of the following reactions produce benzenecarbaldehyde as the major product ?

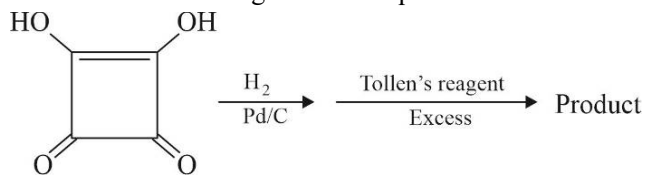


24. Consider the following nucleophilic substitution reaction :



18.3 gm of organic compound (P) reacts with 17.8 gm of organic compound (Q) to form major product (R). Find the mass (in gm) of major product (R). (Given :  $M_0$  of N = 14, O = 16, C = 12, F = 19)

25. Consider the following reaction sequence.



Major product contains 'X' number of lone pairs, 'Y' number of  $sp^2$  hybridized atoms and 'Z' number of  $\sigma$  bonds. Find the sum of X, Y and Z.

SPACE FOR ROUGH WORK

**PART III : MATHEMATICS****100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If plane cuts off intercepts  $OA = a$ ,  $OB = b$ ,  $OC = c$  from the coordinate axes, then the area of the triangle  $ABC$  equal to:
 

(A)  $\frac{1}{2}\sqrt{b^2c^2 + c^2a^2 + a^2b^2}$  (B)  $\frac{1}{2}(bc + ca + ab)$

(C)  $\frac{1}{2}abc$  (D)  $\frac{1}{2}\sqrt{(b+c)^2(c-a)^2 + (a-b)^2}$
- A variable plane passes through a fixed point  $(1, 2, 3)$ . The locus of the foot of the perpendicular drawn from origin to this plane is:
 

(A)  $x^2 + y^2 + z^2 - x - 2y - 3z = 0$  (B)  $x^2 + 2y^2 + 3z^2 - x - 2y - 3z = 0$

(C)  $x^2 + 4y^2 + 9z^2 + x + 2y + 3 = 0$  (D)  $x^2 + y^2 + z^2 + x + 2y + 3z = 0$
- The coplanar points A, B, C, D are  $(2-x, 2, 2)$ ,  $(2, 2-y, 2)$ ,  $(2, 2, 2-z)$  and  $(1, 1, 1)$  respectively. Then:
 

(A)  $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$  (B)  $x + y + z = 1$

(C)  $\frac{1}{1-x} + \frac{1}{1-y} + \frac{1}{1-z} = 1$  (D) None of these
- Two systems of rectangular axes have same origin. If a plane cuts them at distance  $a, b, c$  and  $a_1, b_1, c_1$  from the origin, then:
 

(A)  $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a_1^2} + \frac{1}{b_1^2} + \frac{1}{c_1^2}$  (B)  $\frac{1}{a^2} - \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a_1^2} - \frac{1}{b_1^2} + \frac{1}{c_1^2}$

(C)  $a^2 + b^2 + c^2 = a_1^2 + b_1^2 + c_1^2$  (D)  $a^2 - b^2 + c^2 = a_1^2 + b_1^2 + c_1^2$
- The base of the pyramid AOBC is an equilateral triangle OBA with each side equal to  $4\sqrt{2}$ , 'O' is the origin of reference, AC is perpendicular to the plane of  $\triangle OBC$  and  $|\overline{AC}| = 2$ . Then the cosine of the angle between the skew straight lines one passing through A and the mid-point of OB and the other passing through O and the mid-point of BC is:
 

(A)  $-\frac{1}{\sqrt{2}}$  (B) 0 (C)  $\frac{1}{\sqrt{6}}$  (D)  $\frac{1}{\sqrt{2}}$
- A variable plane is at a constant distance  $p$  from the origin and meets the axes in A, B and C. The locus of the centroid of the tetrahedron OABC is:
 

(A)  $x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$  (B)  $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p}$

(C)  $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = 16$  (D) None of these

7. Minimum value of  $x^2 + y^2 + z^2$  when  $ax + by + cz = p$  is:  
 (A)  $\frac{p}{\Sigma a}$  (B)  $\frac{p^2}{\Sigma a^2}$  (C)  $\frac{\Sigma a^2}{p}$  (D) 0
8. Taken on side  $\overline{AC}$  of a triangle ABC, a point M such that  $\overline{AM} = \frac{1}{3}\overline{AC}$ . A point N is taken on the side  $\overline{CB}$  such that  $\overline{BN} = \overline{CB}$ , then for the point of intersection X of  $\overline{AB}$  and  $\overline{MN}$  which of the following holds good?  
 (A)  $\overline{XB} = \frac{1}{3}\overline{AB}$  (B)  $\overline{AX} = \frac{1}{3}\overline{AB}$  (C)  $\overline{XN} = \frac{3}{4}\overline{MN}$  (D)  $\overline{XM} = 3\overline{XN}$
9. If the unit vectors  $\vec{e}_1$  and  $\vec{e}_2$  are inclined at an angle  $2\theta$  and  $|\vec{e}_1 - \vec{e}_2| < 1$ , then for  $\theta \in [0, \pi]$ ,  $\theta$  may lie in the interval:  
 (A)  $\left[0, \frac{\pi}{6}\right)$  (B)  $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$  (C)  $\left[\frac{\pi}{2}, \frac{5\pi}{6}\right]$  (D) None of these
10. If  $\vec{a}, \vec{b}, \vec{c}$  are three non-coplanar non-zero vectors and  $\vec{r}$  is any vector in space, then  $(\vec{a} \times \vec{b}) \times (\vec{r} \times \vec{c}) + (\vec{b} \times \vec{c}) \times (\vec{r} \times \vec{a}) + (\vec{c} \times \vec{a}) \times (\vec{r} \times \vec{b})$  is equal to:  
 (A)  $2[\vec{a} \vec{b} \vec{c}]\vec{r}$  (B)  $3[\vec{a} \vec{b} \vec{c}]\vec{r}$  (C)  $[\vec{a} \vec{b} \vec{c}]\vec{r}$  (D) None of these
11. Vector  $\vec{x}$  satisfying the relation  $\vec{A} \cdot \vec{x} = c$  and  $\vec{A} \times \vec{x} = \vec{B}$  is:  
 (A)  $\frac{c\vec{A} - (\vec{A} \times \vec{B})}{|\vec{A}|}$  (B)  $\frac{c\vec{A} - (\vec{A} \times \vec{B})}{|\vec{A}|^2}$  (C)  $\frac{c\vec{A} + (\vec{A} \times \vec{B})}{|\vec{A}|^2}$  (D)  $\frac{c\vec{A} - 2(\vec{A} \times \vec{B})}{|\vec{A}|^2}$
12. Urn A contains 9 red balls and 11 white balls. Urn B contains 12 red balls and 3 white balls. A person is to roll a single fair die. If the result is a one or a two, then he is to randomly select a ball from urn A. Otherwise he is to randomly select a ball from urn B. The probability of obtaining a red ball, is:  
 (A)  $\frac{41}{60}$  (B)  $\frac{19}{60}$  (C)  $\frac{21}{35}$  (D)  $\frac{35}{60}$
13. Two coins look similar, but have different probabilities of falling "head". One is a fair coin, with  $P(H) = \frac{1}{2}$ , but the other is weighed so that  $P(H) = \frac{4}{5}$ . One of the coin is chosen at random and is tossed 10 times, let X denotes the number of heads and F is the event that the fair coin was drawn. If  $X = 7$  is observed, the probability that the coin was fair, is:  
 (A)  $\frac{5^{10}}{5^8 + 8^{10}}$  (B)  $\frac{5^8}{5^{10} + 4^{10}}$  (C)  $\frac{5^{10}}{5^{10} + 8^7}$  (D)  $\frac{5^{10}}{5^{10} + 8^8}$
14. A taxi was involved in a hit and run accident at night. There are two taxi companies in the city, namely Black Taxis and White Taxis. It is known that 85% of the taxis in the city are Black and 15% are White. There was a witness to the accident and, according to the witness, the taxi involved in the accident was White. Further investigation of the reliability of the witness showed that, the witness was able to identify correctly the colour of a taxi, 80% of the time. The probability that the taxi involved was Black, is  
 (A)  $\frac{17}{29}$  (B)  $\frac{15}{29}$  (C)  $\frac{12}{29}$  (D)  $\frac{14}{29}$

15. Let A and B are two events of a random experiment such that  $P(A) = \frac{1}{4}$ ,  $P\left(\frac{B}{A}\right) = \frac{1}{2}$  and  $P\left(\frac{A}{B}\right) = \frac{1}{4}$ , then  $P\left(\frac{\bar{A}}{\bar{B}}\right)$  is equal to:
- (A)  $\frac{3}{4}$  (B)  $\frac{1}{4}$  (C)  $\frac{1}{3}$  (D)  $\frac{2}{3}$
16. Let  $A = \{1, 3, 5, 7, 9\}$  and  $B = \{2, 4, 6, 8\}$  be two set. An element  $(a, b)$  of their Cartesian product  $A \times B$  is chosen at random. The probability that  $(a + b) = 9$  is:
- (A)  $\frac{4}{5}$  (B)  $\frac{3}{5}$  (C)  $\frac{2}{5}$  (D)  $\frac{1}{5}$
17. Five different digits from the set of numbers  $\{1, 2, 3, 4, 5, 6, 7\}$  are written in a random order. The probability that 5 digit number thus formed is divisible by 9 is:
- (A)  $\frac{1}{21}$  (B)  $\frac{2}{21}$  (C)  $\frac{5}{21}$  (D)  $\frac{7}{21}$
18. A teacher conducts quiz among the five students of his batch and distributes the answer sheets among them randomly for evaluation then the probability that there are at least two students who are not evaluating their own answer sheet, is equal to:
- (A)  $\frac{1}{120}$  (B)  $\frac{7}{120}$  (C)  $\frac{119}{120}$  (D)  $\frac{113}{120}$
19. If  $a$  and  $b$  are chosen randomly from the set consisting of numbers 1, 2, 3, 4, 5, 6 with replacement, then the probability that  $\lim_{x \rightarrow 0} \left( \frac{a^x + b^x}{2} \right)^{2/x} = 6$  is:
- (A)  $\frac{2}{9}$  (B)  $\frac{1}{9}$  (C)  $\frac{1}{6}$  (D)  $\frac{1}{3}$
20. The probability that the birthdays of six different persons will fall in exactly two calendar months is:
- (A)  $\frac{341}{12^5}$  (B)  $\frac{66}{12^5}$  (C)  $\frac{352}{12^5}$  (D) None of these

SPACE FOR ROUGH WORK

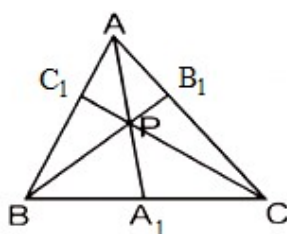
**SECTION-2**

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the  $\oplus$  sign for positive values. However, for negative values,  $\ominus$  sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. The distance of the point  $(1, -2, 3)$  from the plane  $x - y + z = 5$  measured parallel to the line,

$$\frac{x}{2} = \frac{y}{3} = \frac{z}{-6} \text{ is } \underline{\hspace{2cm}}.$$

22. In the adjacent figure 'P' is any arbitrary interior point of the triangle ABC such that the lines  $AA_1, BB_1, CC_1$  are concurrent at P. Value of  $\frac{PA_1}{AA_1} + \frac{PB_1}{BB_1} + \frac{PC_1}{CC_1}$  is always equal to \_\_\_\_.



23. ABC is a triangle where  $A = (2, 3, 5)$ ,  $B = (-1, 2, 2)$  and  $C(\lambda, 5, \mu)$ . If the median through A is equally inclined to all the three positive axes then  $\lambda + \mu$  is \_\_\_\_.
24. If the vectors  $a\hat{i} + \hat{j} + \hat{k}$ ,  $\hat{i} + b\hat{j} + \hat{k}$  and  $\hat{i} + \hat{j} + c\hat{k}$  ( $a \neq b \neq c \neq 1$ ) are coplanar, then the value of  $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$  is equal to \_\_\_\_.
25. The volume of the parallelepiped constructed on the diagonals of the faces of the given rectangular parallelepiped is  $m$  times then volume of the given parallelepiped. Then  $m$  is equal to \_\_\_\_.

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**SPACE FOR ROUGH WORK**